Godiva Leakage Spectrum



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An Unreflected U-235 Critical Assembly[1]

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An unreflected, spherical U^{235} critical assembly has been in operation at the Los Alamos Scientific Laboratory since August, 1951. A remotely controlled mechanical system is used to assemble subcritical components of the sphere, and reactivity is adjusted with U^{235} control rods. The maximum power level during sustained operation is about 1 kw. Investigations with the assembly include studies of the neutron spectrum, observation of the changes of reactivity produced by inserting foreign materials into the assembly, and determination of parameters such as the temperature coefficient of reactivity. In addition, experiments at reactivities above prompt critical have been carried out. The assembly has also been used as a source of short, high-intensity bursts of neutrons in the study of delayed neutrons following fission.

INTRODUCTION

Until 1951, studies of all-metal critical assemblies carried on with the Pajarito Site remotely controlled facilities of the Los Alamos Scientific Laboratory (*1*) were concentrated on configurations of U^{235} surrounded by various reflectors. With these critical assemblies, spatial dependence of fission rates, relations between critical mass and reflector thickness and material, changes in reactivity produced by the introduction of nonfissionable materials into the assemblies, distribution-in-energy of neutrons in the fissionable material, and time-dependent behavior of prompt neutron fission chains were investigated.

An extension of these studies with a bare assembly consisting entirely of fissionable material in a simple geometry seemed desirable, since the number of parameters affecting the interpretation of measurements would be reduced. Accordingly, a nearly spherical and unreflected assembly of uranium, enriched to a U^{235} content of approximately 90%,[3] was constructed. This device, known as "Lady Godiva," was first operated at delayed critical in August, 1951.

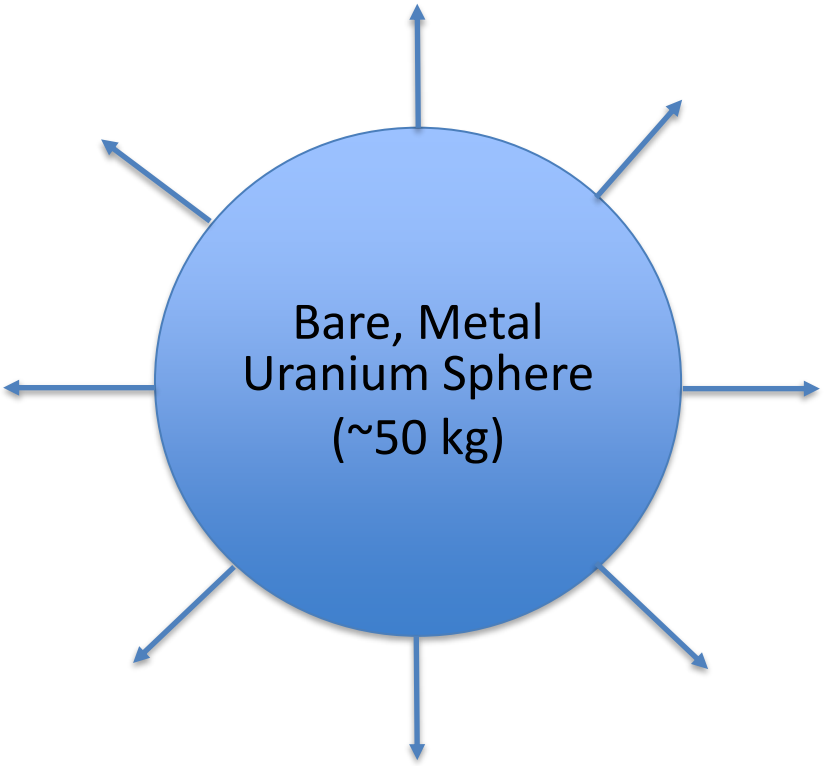
This paper deals mainly with details of the design and with parameters of

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[3] In subsequent references, the term "uranium" implies material of this enrichment in U^{235}.

Godiva Test Assembly



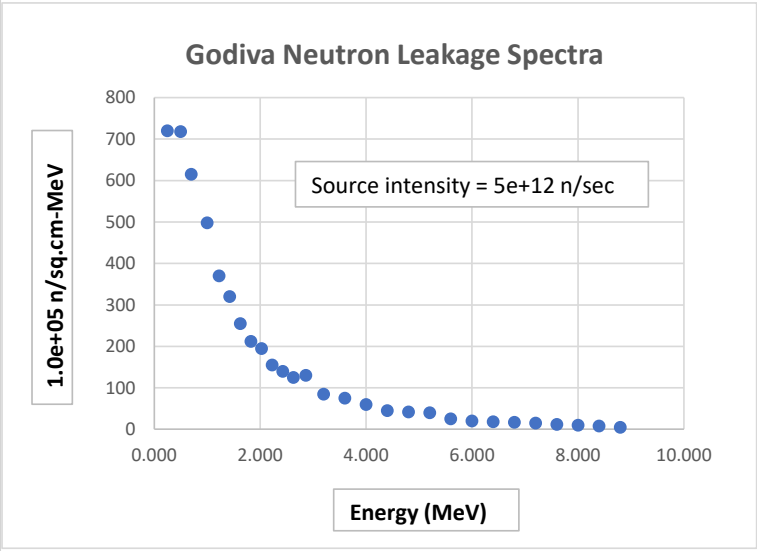
The Godiva[5;6] experimental configuration is the ^{235}U analog to Jezebel. It is a bare, spherical, high-enriched (93.0% ^{235}U, 6.0% ^{238}U, 1.0% ^{234}U) metal system of radius 8.7313 cm with a metal density of 19.0 g/cc. The experimental configuration is similar to that of Jezebel with a steel coating and nominal steel structural supports.

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Measured Godiva Leakage Spectra

Energy (MeV)	N(E) (1e+05 n/sq.cm-MeV)	E1	E2	ΔE
0.250	720	0.125	0.375	0.250
0.500	718	0.375	0.625	0.250
0.700	615	0.625	0.875	0.250
1.000	498	0.875	1.125	0.250
1.225	370	1.125	1.325	0.200
1.425	320	1.325	1.525	0.200
1.626	255	1.525	1.725	0.200
1.825	212	1.725	1.925	0.200
2.025	195	1.925	2.125	0.200
2.225	155	2.125	2.325	0.200
2.425	140	2.325	2.525	0.200
2.625	125	2.525	2.725	0.200
2.863	130	2.725	3.000	0.275
3.200	85	3.000	3.400	0.400
3.600	75	3.400	3.800	0.400
4.000	60	3.800	4.200	0.400
4.400	45	4.200	4.600	0.400
4.800	42	4.600	5.000	0.400
5.200	40	5.000	5.400	0.400
5.600	25	5.400	5.800	0.400
6.000	20	5.800	6.200	0.400
6.400	18	6.200	6.600	0.400
6.800	17	6.600	7.000	0.400
7.200	15	7.000	7.400	0.400
7.600	12	7.400	7.800	0.400
8.000	10	7.800	8.200	0.400
8.400	8	8.200	8.600	0.400
8.800	5	8.600	9.000	0.400

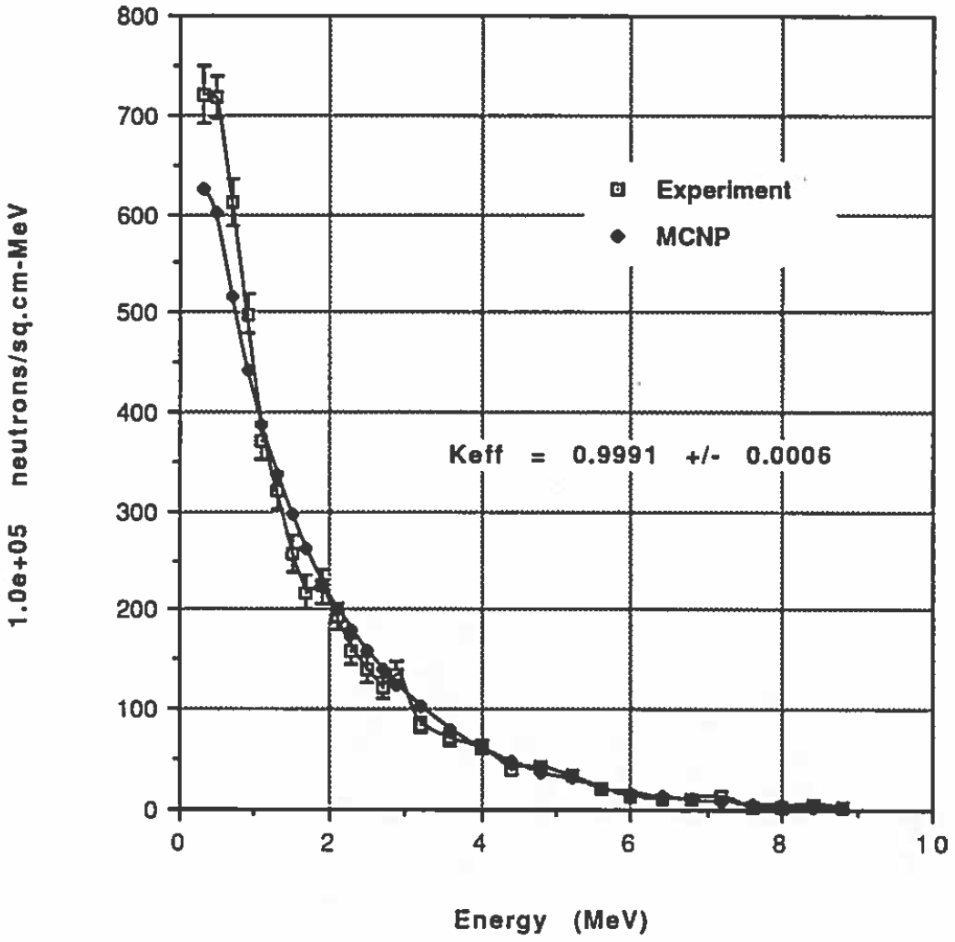
Peterson, R.E., Newby, G.A., "An Unreflected U^{235} Critical Assembly", *Nuclear Science & Engineering*, Vol. 1, pp. 112-125 (1956)



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Comparison to MCNP

Ref. (WSRC-MS-92-408)



At 100 watts, the neutron leakage from Godiva is about 5×10^{12} n/sec.

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Computational Problem

- Use MCNP spherical model of Godiva assembly provided
- Utilize material compositions and geometry provided
- Model 50 cm air region surrounding metal sphere
- Tally neutron leakage flux coming off surface assembly
- Utilize flux energy binning structure provided (ΔE values) to convert to n/sq-cm-MeV
- Normalize tallies to experimental data (you will need to divide final MCNP results by 100 to correct for power level)
- Compare measured and MCNP calculated leakage spectra
- Confirm critical eigenvalue (k_{eff} ~ 1.0)
- *Note: You can ignore the first MCNP tally result from 0.0 – 0.125 MeV when generating your plot since this calculated data point does not correspond to anything physical in the experiment*

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Example MCNP Input File

```
      Godiva Neutron Leakage Spectra Problem
1     1 0.0486446      -1  imp:n=1
2     2 0.0000538    1 -2  imp:n=1
3     0                2  imp:n=0

1     so  8.7313
2     so  50.0

kcode 10000 1.0 200 2000
ksrc  0.0 0.0 0.0
f12:n 1    $ Surface Flux Tally for Neutrons
e12   0.125 0.375 0.625 0.875 1.125 1.325 1.525 1.725 1.925 2.125 2.325
      2.525 2.725 3.0 3.4 3.8 4.2 4.6 5.0 5.4 5.8 6.2 6.6 7.0 7.4 7.8
      8.2 8.6 9.0
m1    92234.50c  0.00048888  $ U(93%) Metal
      92235.50c  0.04527185
      92238.50c  0.00288387
m2    7014.50c   0.0000425   $ Air at STP
      8016.50c   0.0000113
```

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